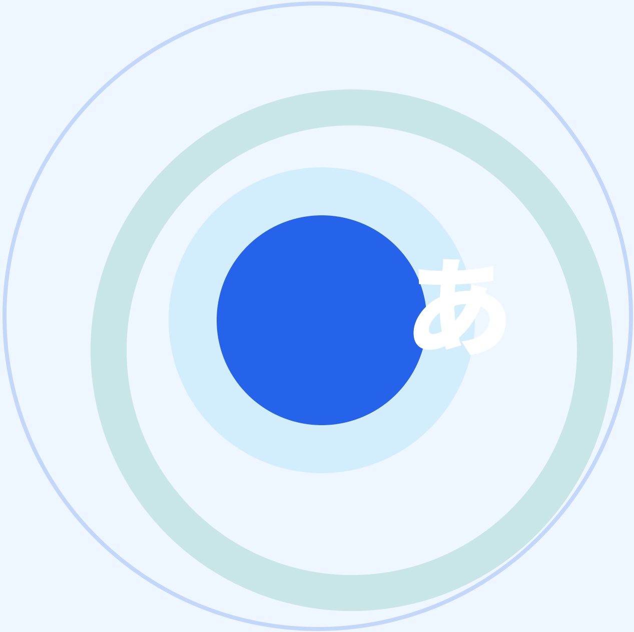
MASTER OF SOFTWARE ENGINEERING - THESIS DEFENSE

VAJA: Personalized Japanese Learning Assistant

Design and implementation of a web-based learning assistant using role-based learning agents, Knowledge Graph, RAG, personalized pathways, flashcards, and assessment for N5/N4 learners.



Student
Bui Duy Hieu

Class
MSE K24

Supervisor
Dr. Nguyen Duy Nghiem

PROBLEM

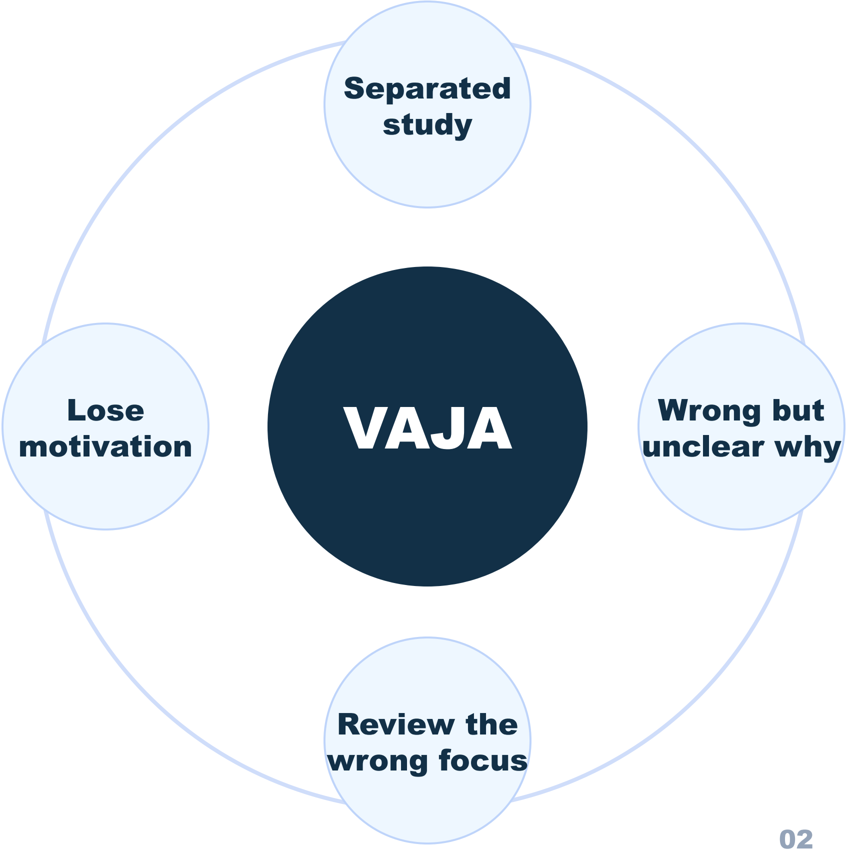
Beginner learners can easily get lost when studying alone

For N5/N4, learners need a clear study flow, grounded answers, and review tasks based on their weak points.

1 **Unclear next step**
Many learners study from separated lists without a clear pathway.

2 **Chatbot answers lack sources**
LLMs can explain well, but they do not always follow N5/N4 learning material.

3 **Review is not focused**
Quiz, flashcards, and feedback are often not connected into one learning state.



PRESENTATION FLOW

Agenda

The presentation follows six parts and focuses on system design, implementation, and evaluation from an MSE view.

01

N5/N4 study problem

New learners can get lost because content is separated and sources are unclear.

02

Agent scope

Agents are role-based services, not a fully autonomous multi-agent system.

03

System architecture

React, Python FastAPI backend, PostgreSQL, Neo4j, and Qdrant.

04

Personalized learning

Based on profile, pathway, quiz, flashcards, feedback, and weak skills.

05

Measured results

RAG benchmark, pilot test, SUS/trust, pass rate, and test coverage.

06

Demo and limits

Show the running app, what was achieved, and what still needs improvement.

RESEARCH QUESTIONS

5 main questions

The questions match the thesis evaluation plan: KG, RAG, mastery, role state, and pilot usability.

RQ1

How is the KG built?

Store N5/N4 words, grammar, kanji, lessons, and study links in Neo4j.

RQ2

Does KG + Vector RAG help?

Compare LLM only, Vector, KG, and KG + Vector using retrieval metrics.

RQ3

How does mastery change tasks?

Use quiz and flashcard signals to slow down, review, or move faster.

RQ4

How do roles share state?

Tutor, Pathway, Assessment, and Review share learner state without mixing jobs.

RQ5

Can learners use it?

Pilot test with Vietnamese learners: SUS, trust, scores, feedback, and comments.

SCOPE CLARIFICATION

Role-based learning agent scope

Each role has a clear job, input, output, and shared learner state. This is not autonomous multi-agent AI.

TERM IN THESIS	REAL IMPLEMENTATION	NOT CLAIMED
Tutor role	Retrieves KG/vector sources and asks the LLM to answer.	No autonomous goal setting.
Pathway role	Reads profile, scores, weak items, due cards, and study time.	No BKT/DKT/IRT model in the current MVP.
Assessment role	Grades quiz/test answers and records mastery signals.	No adaptive testing model yet.
Review role	Uses SM-2 style flashcard ratings and next-review time.	No independent planning agent.

RESEARCH SCOPE

What this thesis does not claim

These limits are stated clearly so the evaluation is interpreted in the right scope.

!

No autonomous multi-agent claim

Agents mean role-based learning services with shared learner state, not fully autonomous negotiation or planning agents.

!

No BKT, DKT, or IRT claim

The current pathway uses heuristic rules and learning signals. Formal mastery models are future work.

!

No proven learning gain claim

The pilot checks usability and flow. It does not prove long-term improvement in Japanese ability.

!

No always-correct LLM claim

RAG provides sources and grounding, but answer correctness and faithfulness still need separate human evaluation.

!

No population-wide pilot claim

The pilot is small and local. It is not representative of all Vietnamese Japanese learners.

Overall architecture

Thesis architecture: React frontend, Python FastAPI backend, PostgreSQL, Neo4j, and Qdrant.

Frontend

- **React + Vite**
- **Learner UI**
- **Floating Tutor**
- **Responsive Web**

FastAPI Backend

- **Auth + JWT + Refresh Token**
- **Profile + Progress + Metrics**
- **Tutor + Planner + Assessment APIs**
- **API Contract + Security**

Data + Retrieval

- **PostgreSQL Learner Data**
- **Neo4j Knowledge Graph**
- **Qdrant Vector Search**
- **LLM Provider**

Main learning flow

The learner follows an unlocked pathway, similar to short learning chapters.

1 Onboarding

8 questions about level, goal, time, and weak points.

2 Pathway

Each chapter has about 3 lessons and unlocks by result.

3 Flashcards

Study words, patterns, kana, and kanji before the quiz.

4 Lesson quiz

Score 85% or higher to unlock the next lesson.

5 Chapter test

20 questions; 85% is required for the next chapter.

6 Feedback

Store learner feeling to adjust the pace.

Personalized pathway

The pathway changes based on real learning data, not only simple UI branching.

Input signals

- Current level: Zero, N5, N4
- Learning pathway: JLPT, conversation, school, work, reading
- Daily study minutes
- Weak skills: kana, grammar, vocab, kanji, listening, speaking
- Quiz score, attempts, fail count, feedback

Adaptive output

- Zero beginners study kana first
- Many failures: stay, review mistakes, ask Tutor
- Good results: faster pace, unlock next lesson
- End of chapter: required 20-question test
- Hard/easy feedback is stored as a pilot signal

AI Tutor answers with sources

The Tutor uses lesson context, retrieves sources before answering, and stores evidence.

Tutor flow

- Receive question + contextTopic
- Search Knowledge Graph and Vector DB
- Call LLM to explain in Vietnamese
- Return answer, sources, and confidence

Learning safety rules

- No source means no exposure record
- Chat does not increase mastery
- Quiz/Assessment records correct or wrong
- Flashcard review records Again, Hard, Good, Easy

Controlled learning assessment

Quiz and chapter test give clearer learning signals than chat alone.

Lesson quiz

5 short questions. 85% is required to unlock the next lesson.

Chapter test

20 mixed questions. 85% is required for the next chapter.

Answer key

Stored in backend. Start session does not return answers.

Learning signal

QUIZ and ASSESSMENT only accept CORRECT or WRONG.

Flashcard SRS

SM-2 schedule with Again, Hard, Good, Easy.

Feedback

Stores difficulty, pace, and review or move-on choice.

Software engineering contribution

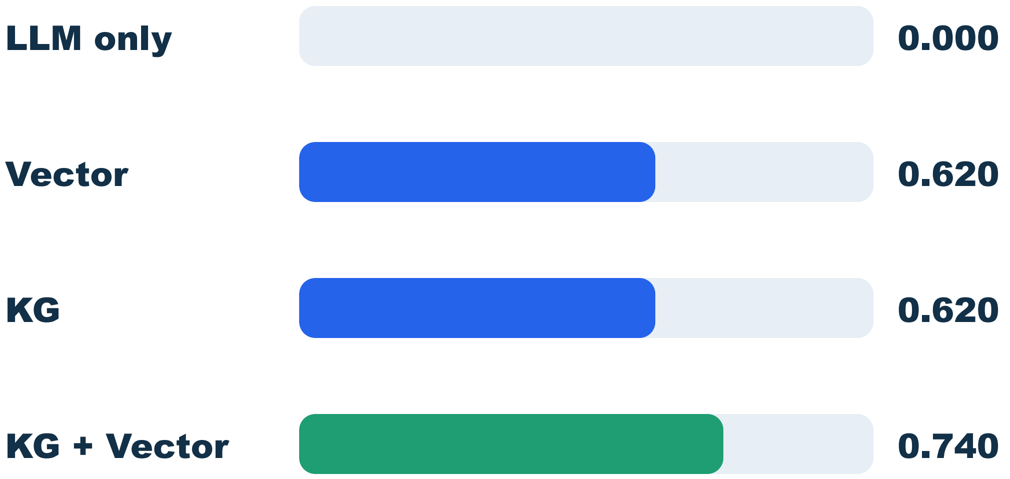
The focus is a system with clear architecture, testability, and deployability.

- ✓ **Clean architecture**
  Thin controllers, service interfaces, request/response DTOs, separate exception handler.
- ✓ **Clear API contract**
  Auth, profile, chat, flashcards, assessment, feedback, metrics, planner.
- ✓ **Security**
  JWT access token, refresh token, Google OAuth linking, BCrypt password.
- ✓ **Reproducibility**
  Docker Compose, KG seed, benchmark script, automated tests.
- ✓ **Deployment**
  Frontend Netlify, backend VPS, proxy API, public demo link.

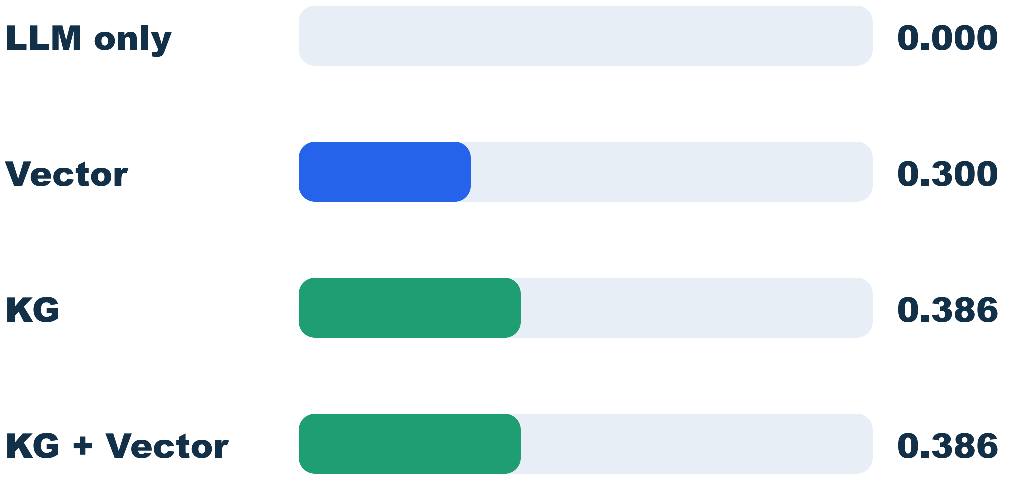
Retrieval quality result

50 N5/N4 questions, 4 modes. These metrics measure retrieved sources, not final answer correctness.

Source Recall



Precision@3



Important: Source Recall 0.740 is not answer accuracy 74%. Answer correctness and faithfulness require a separate evaluation.

Pilot sample consistency

Different endpoints produce different counts. They are related logs, not one combined sample size.

Pilot learners

10

Main learner group used for lesson-flow observation.

Survey rows

11

SUS/trust submissions; counted as rows, not unique learners.

Feedback users

11

Distinct users who submitted in-app study feedback.

Progress users

19

Users with knowledge-progress records, including pilot/demo/test use.

Lesson attempts

25

Attempt logs, not unique learner count.

Avg score

82.4%

Average over recorded lesson attempts.

Pass rate

56%

14 passed attempts out of 25 attempts.

SUS / Trust

58.6

Trust average: 3.45/5. Moderate, not high.

Improvements after user feedback

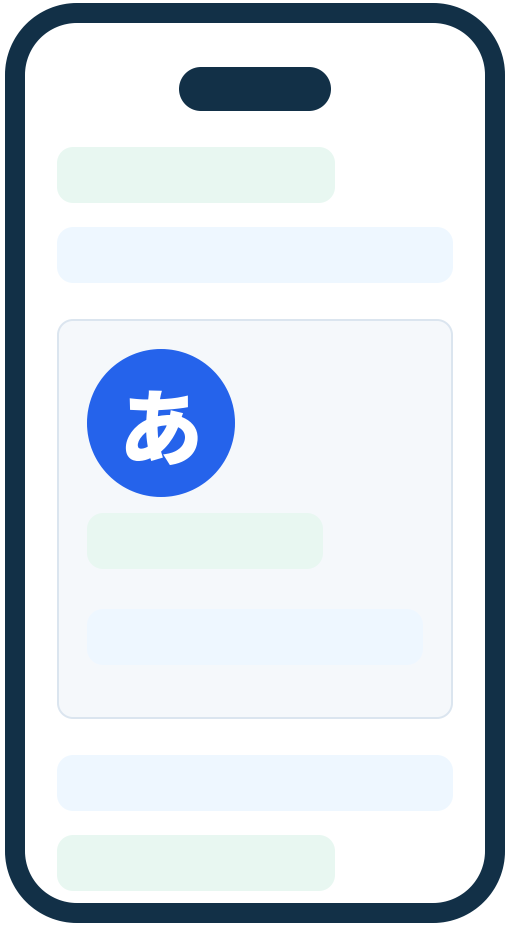
These changes came from real feedback and end-user testing.

Feedback	App change	Purpose
New users did not see the big picture	Added tour guide and kana overview	Easier start from zero
The learning flow felt separated	Chapter pathway, 3 lessons per chapter	Clear study progress
Assessment had no clear role	20-question chapter test, 85% required	Real learning checkpoint
Missing audio/pronunciation	Flashcard audio and pronunciation guidance	Better support for communication
Chatbot should not be a separate tab	Floating Tutor supports quiz context	Ask when needed, without breaking flow

DEMO PLAN

Short demo scenario

The demo focuses on a zero-beginner learner.



1. **Landing → Onboarding**
   Choose new learner and answer 8 questions.
2. **Zero beginner**
   See create-account option or 3-lesson trial.
3. **Study Kana**
   Use cards, listen to audio, finish lesson quiz.
4. **Chapter gate**
   After 3 lessons, take a 20-question chapter test.
5. **Tutor and tools**
   Ask VAJA, use lookup, review flashcards.

Conclusion and future work

VAJA reached the prototype goal and collected initial evaluation data.

Achieved

- End-to-end system works
- Clearer pathway for new learners
- Grounded Tutor with RAG sources
- Retrieval benchmark and pilot data for appendix
- Automated tests: frontend, backend, and retrieval modules

Limitations and next steps

- Pilot is small; results are not population-wide
- Pathway still uses heuristic, not BKT/IRT yet
- Answer quality and faithfulness need separate evaluation
- Need speech recognition and real pronunciation data